Module 4: Classification

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Introduction to Classification

Classification

- Qualitative variables take values in an unordered set $\mathcal{C}$, such as:
  eye color $\in \{\text{brown, blue, green}\}$
  email $\in \{\text{spam, ham}\}$
- Given a feature vector X and a qualitative response Y taking values in the set $\mathcal{C}$, the classification task is to build a function $G(X)$ that takes as input the feature vector X and predicts its value for Y, i.e. $G(X) \in C$.
- Often we are more interested in estimating the probabilities that X belongs to each category in $\mathcal{C}$.
  For example, it is more valuable to have an estimate of the probability that an insurance claim is fraudulent, than a classification fraudulent or not.

Classification

- Assume $y \in \{1, \ldots, K\}$ and $\mathbf{x} \in \mathbb{R}^p$

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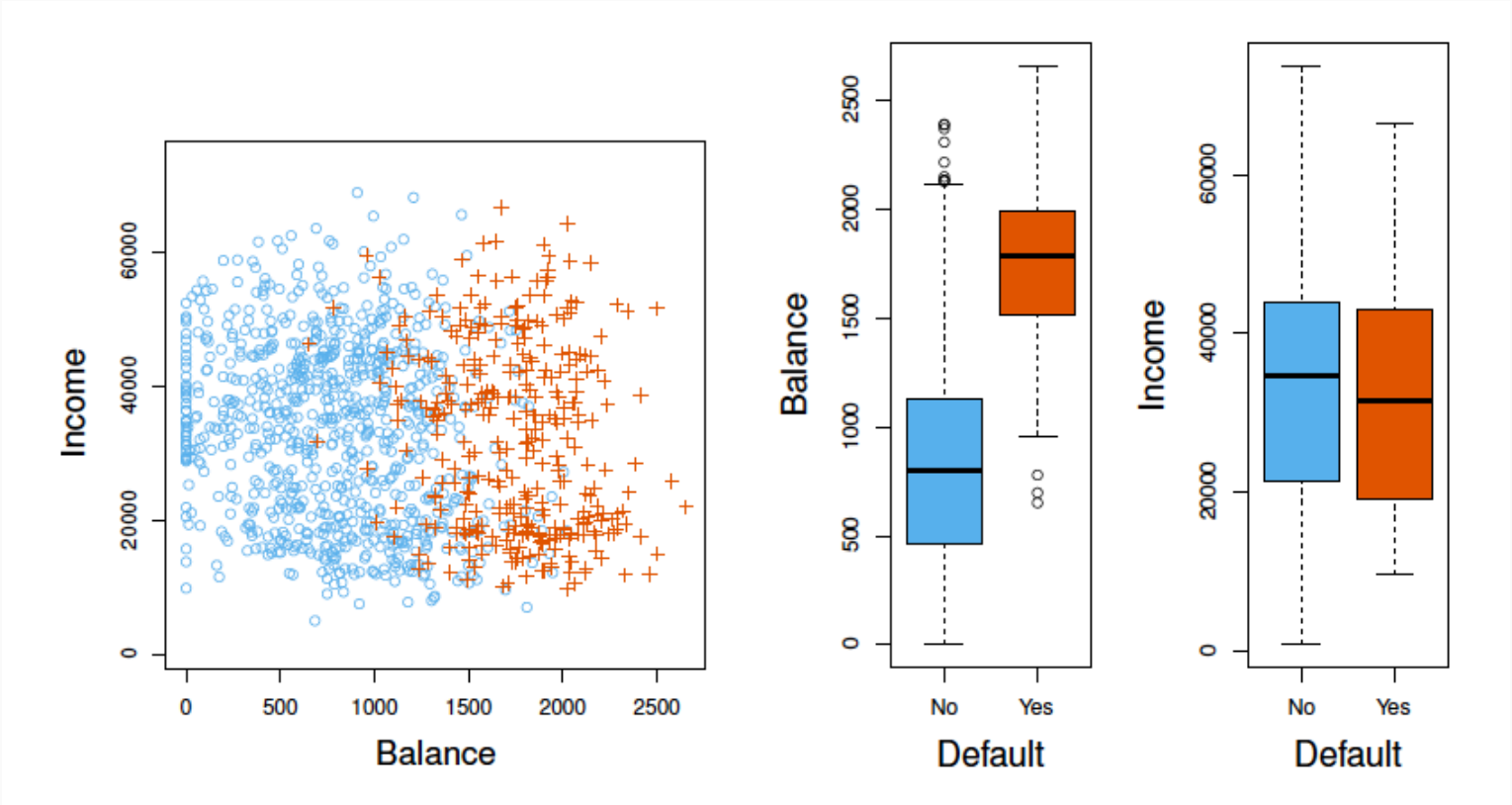
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- Most ML methods attempt to estimate $G^*(\mathbf{x})$ directly
- Yet deep learning methods come back to $Pr(y = k|\, \mathbf{x})$ ...

Credit Card Default



Can we use Linear Regression?

- Suppose for the Default classification task that we code

$$Y = \begin{cases} 0 & \text{if No} \\ 1 & \text{if Yes} \end{cases}$$

Can we simple perform a linear regression of Y on X and classify as *Yes* if $\hat{Y} > 0.5$?

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- Since in the population $\mathbb{E}(Y|X = x) = Pr(Y = 1|X = x)$, we might think that regression is perfect for this task.
- However, linear regression might produce probabilities less than zero or bigger than one. Logistic regression is more appropriate.

Linear Classifiers

Some concepts:

- Linear classification function $f_k(\mathbf{x}) = \beta_{k0} + \mathbf{x}^T \beta_k$
- Classifier $G(\mathbf{x})$ is

$$G(\mathbf{x}) = \underset{k}{\operatorname{argmax}}\, f_k(\mathbf{x})$$

- Decision boundary between classes k and l:

$$\{\mathbf{x} : f_k(\mathbf{x}) = f_l(\mathbf{x})\}$$

- Estimate $f_k(\mathbf{x})$ based on the available training data, leading the estimated $\widehat{G}(\mathbf{x})$

Indicator response matrix:

$$y = \begin{pmatrix} 3 \\ 2 \\ 4 \\ \vdots \\ 1 \end{pmatrix} \Longrightarrow \mathbf{Y} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

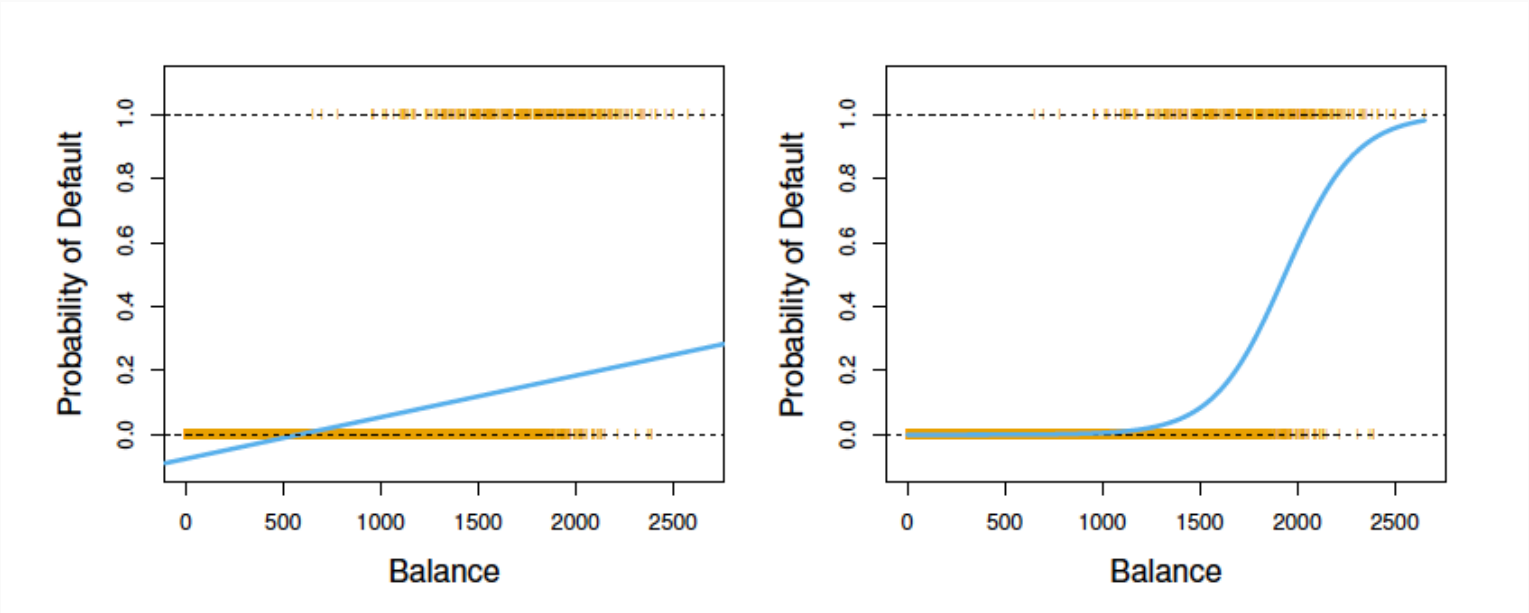
Then the solution is

$$\widehat{\mathbf{B}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$$

$$
\begin{aligned}
\hat{\mathbf{f}}(\mathbf{x}) &= \mathbf{x}^T \widehat{\mathbf{B}} = \begin{pmatrix} \hat{f}_1(\mathbf{x}) \\ \hat{f}_2(\mathbf{x}) \\ \vdots \\ \hat{f}_K(\mathbf{x}) \end{pmatrix} \\
\hat{G}(\mathbf{x}) &= \underset{k}{\operatorname{argmax}}\, \hat{f}_k(\mathbf{x})
\end{aligned}
$$

Rationale: it tries to estimate the minimizer of $\mathbb{E}\| \mathbf{y} - \mathbf{f}(\mathbf{x})\|^2$; i.e. $\mathbb{E}(\mathbf{y} \mid \mathbf{x}) = (p_1(\mathbf{x}), \ldots, p_K(\mathbf{x}))^T$. Yet the estimation is restricted to the linear functional space, so $f_k(x)$ can be less than 0 and greater than 1, although $\mathbf{1}^T \hat{\mathbf{f}}(\mathbf{x}) = 0$ for any $\mathbf{x}$.

Linear vs. Logistic regression



The orange marks indicate the response Y, either 0 or 1. Linear regression does not estimate $Pr(Y = 1|X)$ well. Logistic regression seems well suited to the task.

Linear regression continued

Now suppose we have a response variable with three possible values. A patient presents at the emergency room, and we must classify them according to their symptoms.

$$Y = \begin{cases} 1 & \text{if stroke} \\ 2 & \text{if drug overdose} \\ 3 & \text{if epileptic seizure} \end{cases}$$

This coding suggests an ordering, and in fact implies that the difference between stroke and drug overdose is the same as between drug overdose and epileptic seizure.

Linear regression is not appropriate here. Multiclass Logistic Regression or Discriminant Analysis is more appropriate.

Masking problems with linear regression

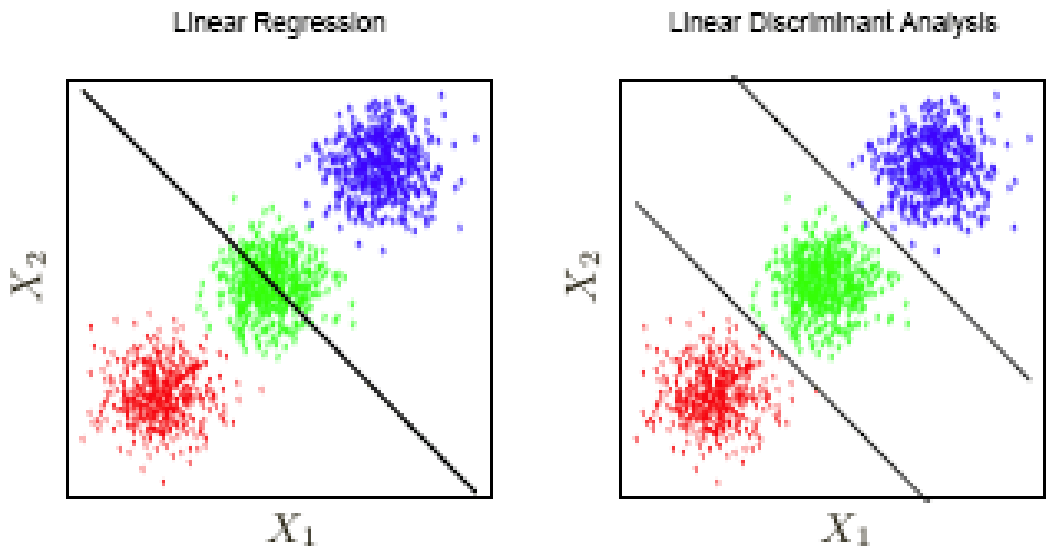
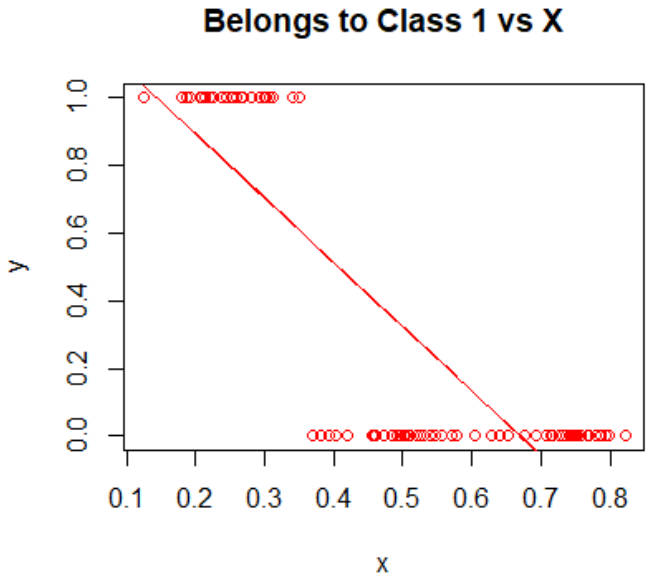
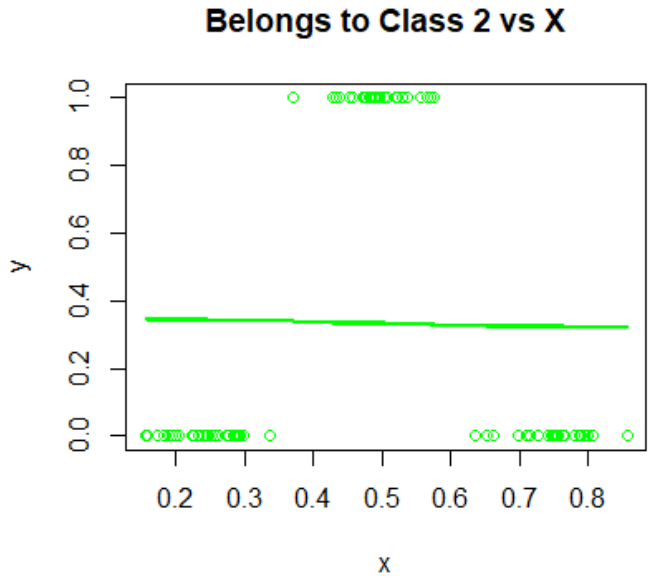


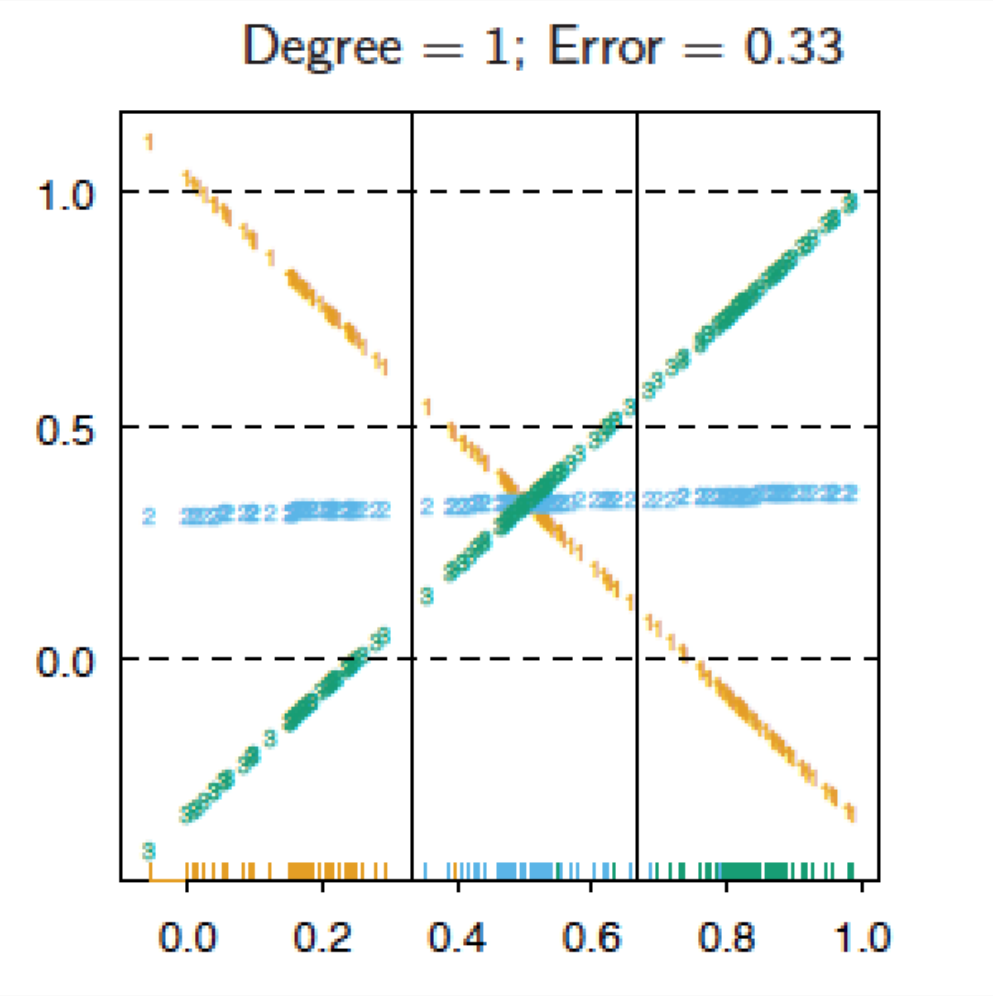
Figure 4.2: *The data come from three classes in* $\mathbb{R}^2$ *and are easily separated by linear decision boundaries. The right plot shows the boundaries found by linear discriminant analysis. The left plot shows the boundaries found by linear regression of the indicator response variables. The middle class is completely masked (never dominates).*

Masking problems with linear regression





Masking problems with linear regression



Discriminant Analysis

Discriminant Analysis

Here the approach is to model the distribution of X in each of the classes separately, and then use Bayes theorem to flip things around and obtain $Pr(Y|X)$.

When we use normal (Gaussian) distributions for each class, this leads to linear or quadratic discriminant analysis.

However, this approach is quite general, and other distributions can be used as well. We will focus on normal distributions.

Bayes theorem for classification

Thomas Bayes was a famous mathematician whose name represents a big subfield of statistical and probabilistic modeling. Here we focus on a simple result, known as Bayes theorem:

$$Pr(Y = k|X = x) = \frac{Pr(X = x|Y = k) \cdot Pr(Y = k)}{Pr(X = x)}$$

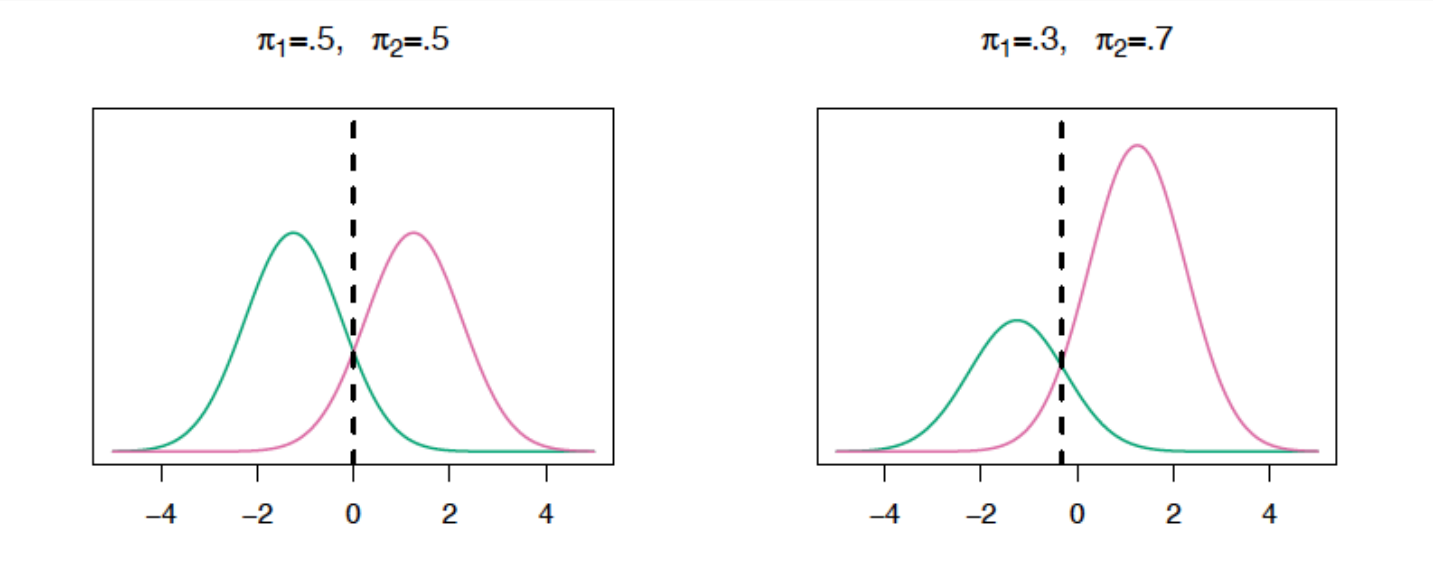
Discriminant analysis

- $f_k(\mathbf{x}) =$ conditional density of $\mathbf{x}$ in class $y = k$
- $\pi_k = Pr(y = k)$ is the class prior
- Bayes theorem

$$p_k(\mathbf{x}) = Pr(y = k|\,\mathbf{x}) = \frac{\pi_k f_k(\mathbf{x})}{\sum_l \pi_l f_l(\mathbf{x})}$$

- Comparing $p_k(\mathbf{x})$ can be simplified to comparing $\pi_k f_k(\mathbf{x})$
- Various assumptions on $f_k(\mathbf{x})$ lead to LDA, QDA, Kernel DA, Naive Bayes, ...

Classify to the highest density



We classify a new point according to which density is highest.

When the priors are different, we take them into account as well, and compare $\pi_k f_k(x)$. On the right, we favor the pink class — the decision boundary has shifted to the left.

Why discriminant analysis?

- When the classes are well-separated, the parameter estimates for the logistic regression model are surprisingly unstable. Linear discriminant analysis does not suffer from this problem.
- If n is small and the distribution of the predictors X is approximately normal in each of the classes, the linear discriminant model is again more stable than the logistic regression model.
- Linear discriminant analysis is popular when we have more than two response classes, because it also provides low-dimensional views of the data.

LDA when $p = 1$

The Gaussian density has the form

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} e^{-\frac{(x-\mu_k)^2}{2\sigma_k^2}}$$

We will assume that all the $\sigma_k = \sigma$ are the same.

Plugging this into Bayes formula, we get a rather complex expression for $p_k(x) = Pr(Y = k|X = x)$:

$$p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu_k)^2}{2\sigma^2}}}{\sum_{l=1}^{K} \pi_l \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu_l)^2}{2\sigma^2}}}$$

Happily, there are simplifications and cancellations.

Discriminant functions

To classify at the value $X = x$, we need to see which of the $p_k(x)$ is largest. Taking logs, and discarding terms that do not depend on k, we see that this is equivalent to assigning x to the class with the largest discriminant score:

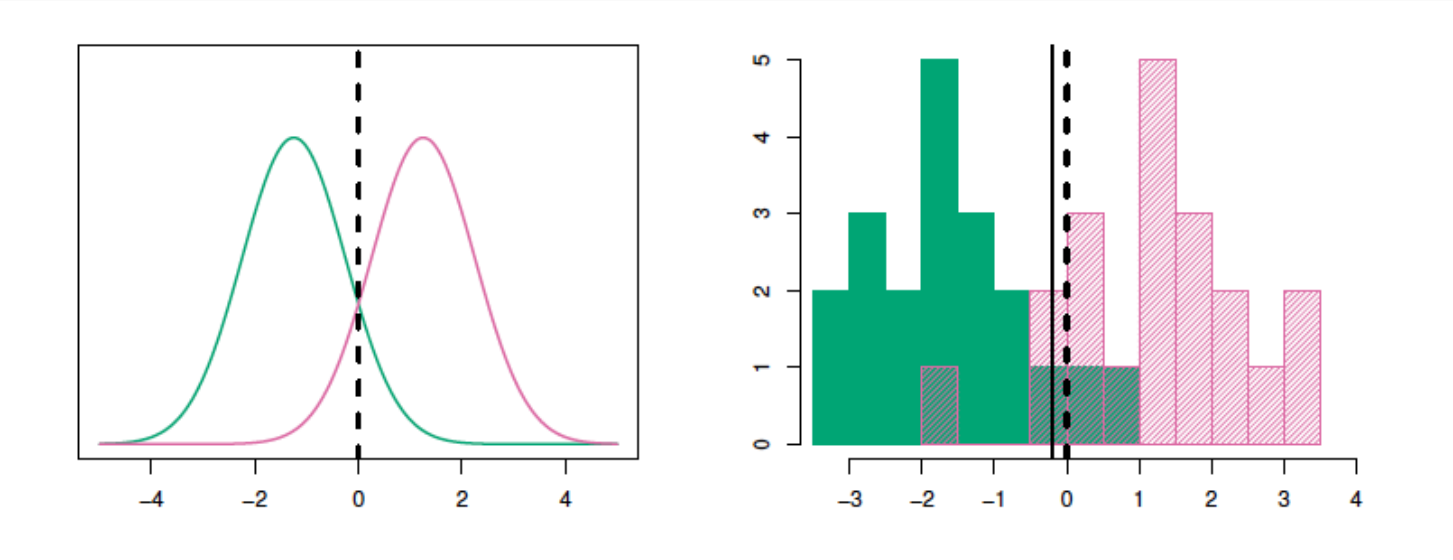
$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

Note that $\delta_k(x)$ is a linear function of x.

If there are $K = 2$ classes and $\pi_1 = \pi_2 = 0.5$, then one can see that the decision boundary is at

$$x = \frac{\mu_1 + \mu_2}{2}$$

(See if you can show this)



Example with $\mu_1 = -1.5$, $\mu_2 = 1.5$, $\pi_1 = \pi_2 = 0.5$, and $\sigma^2 = 1$.

Typically we don't know these parameters; we just have the training data. In that case we simply estimate the parameters and plug them into the rule.

Estimating the parameters

$$\hat{\pi}_k = \frac{n_k}{N}$$

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i$$

$$\hat{\sigma}^2 = \sum_{k=1}^{N} \frac{n_k - 1}{n - K} \hat{\sigma}_k^2$$

where

$$\hat{\sigma}_k^2 = \frac{1}{n_k - 1} \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2$$

is the usual formula for the estimated variance in the kth class.

Linear discriminant analysis (LDA)

- Assume $\mathbf{x}|y = k \sim N_p(\mu_k, \Sigma)$ for $k = 1, \ldots, K$, then

$$f_k(\mathbf{x}) = \frac{1}{(2\pi)^{1/2}|\Sigma|^{1/2}} \exp\{-(1/2)(\mathbf{x} - \mu_k)^T \Sigma^{-1}(\mathbf{x} - \mu_k)\}.$$

  To compare $p_k(\mathbf{x})$'s, we have

$$\begin{aligned} \log \frac{p_k(\mathbf{x})}{p_l(\mathbf{x})} &= \log(\pi_k/\pi_l) - (1/2)(\mu_k + \mu_l)^T \Sigma^{-1}(\mu_k - \mu_l) \\ &\quad + \mathbf{x}^T \Sigma^{-1}(\mu_k - \mu_l), \end{aligned}$$

  which is a linear function in $\mathbf{x}$.

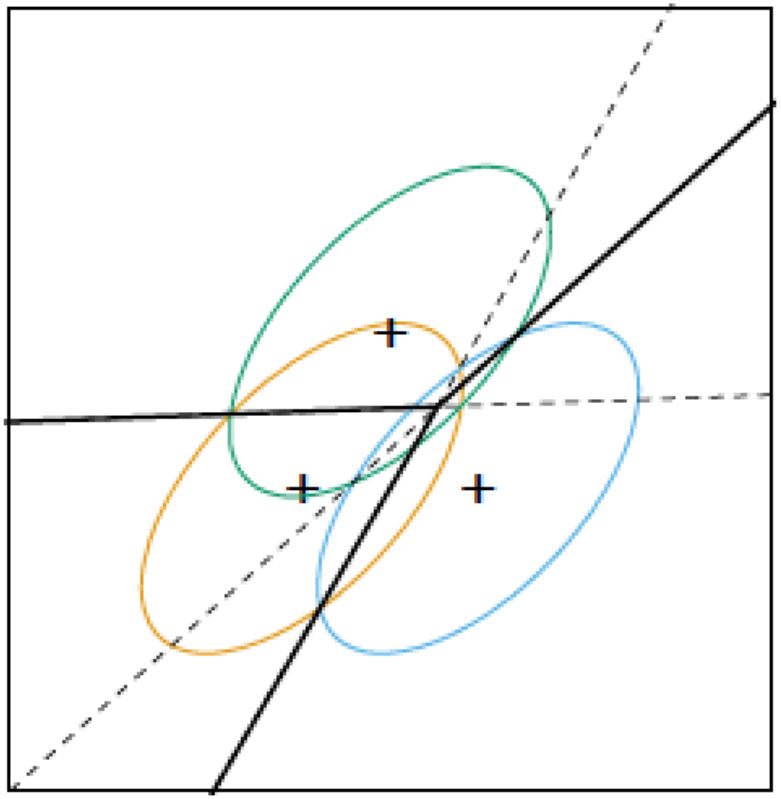
More on LDA

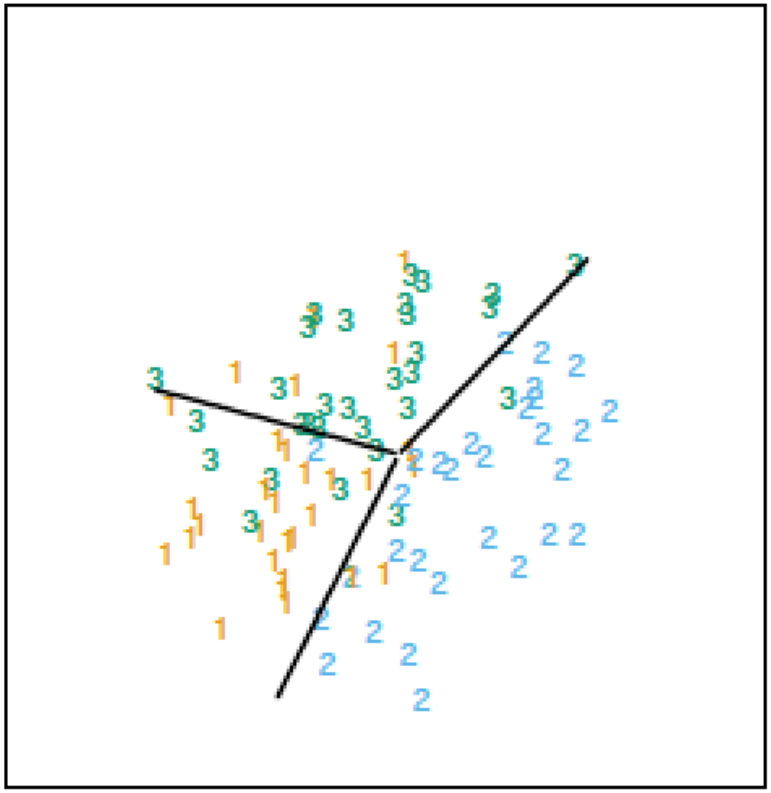
- The quadratic terms of $\mathbf{x}$ vanish because of the equal covariance assumption across classes
- If π_k's are unavailable, $\hat{\pi}_k = n_k/n$
- Estimate μ_k by centroid in class k, and Σ by pooled within class covariance matrix
- $\log \frac{p_k(\mathbf{x})}{p_l(\mathbf{x})} = \delta_k(\mathbf{x}) - \delta_l(\mathbf{x})$, where the discriminant function is

$$\delta_k(\mathbf{x}) = \log(\pi_k) - (1/2)\mu_k^T \Sigma^{-1} \mu_k + \mathbf{x}^T \Sigma^{-1} \mu_k$$

- Classify $\mathbf{x}$ to class k with the largest $\delta_k(\mathbf{x})$

An illustrative example



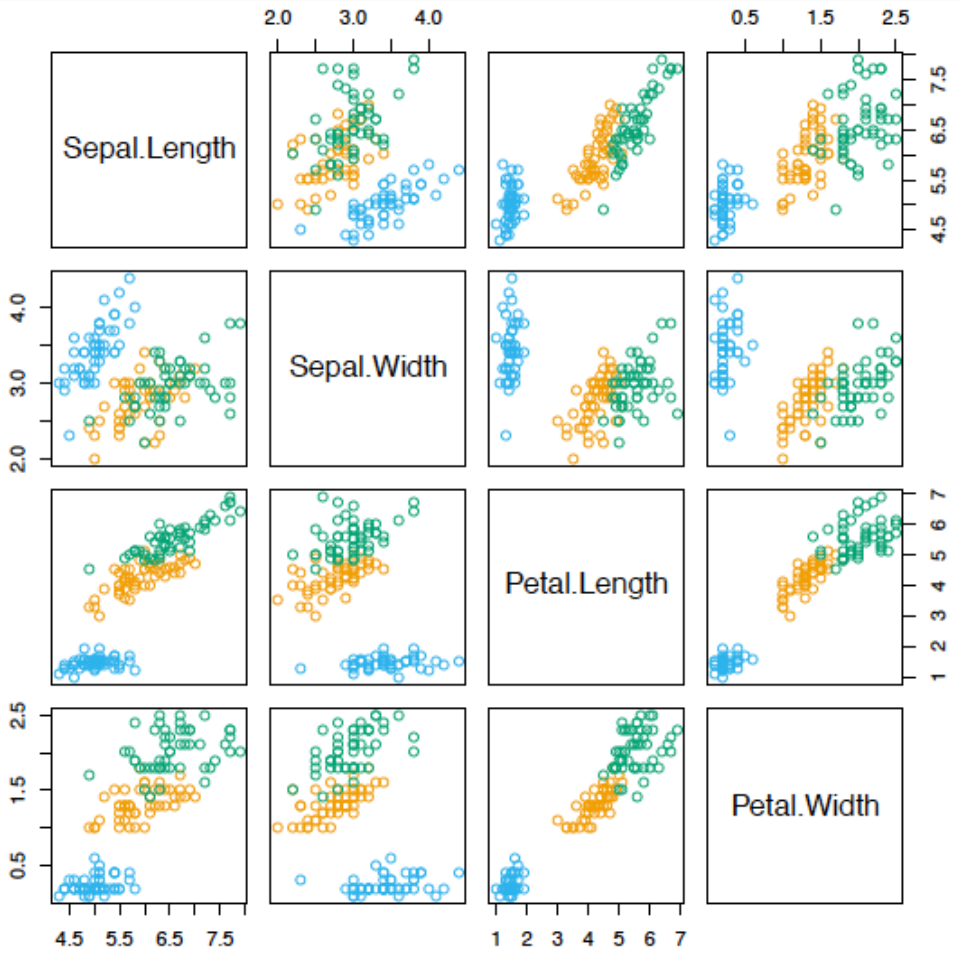


Fisher's Iris Data

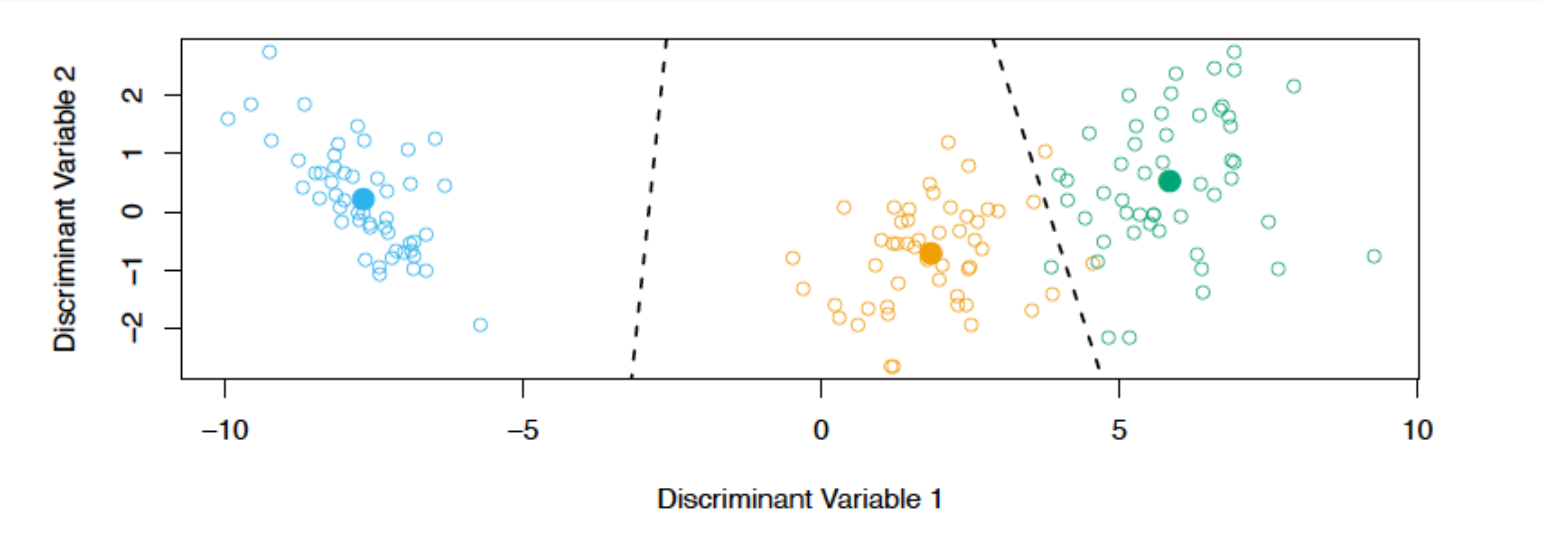
4 variables
3 species
50 samples/class

- Setosa
- Versicolor
- Virginica

LDA classifies all but 3 of the 150 training samples correctly.



Fisher's discriminant plot



When there are K classes, linear discriminant analysis can be viewed exactly in a $K-1$ dimensional plot.

Why? Because it essentially classifies to the closest centroid, and they span a $K-1$ dimensional plane.

Even when $K>3$, we can find the "best" 2-dimensional plane for visualizing the discriminant rule.

LDA on credit data

		True Default Status		
		No	Yes	Total
Predicted	No	9644	252	9896
Default Status	Yes	23	81	104
	Total	9667	333	10000

$(23 + 252)/10000$ errors — 2.75% misclassification rate!

Some caveats:

- This is training error, and we may be overfitting.

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- If we classified to the prior — always to class No in this case — we would make 333/10000 errors, or only 3.33%.
- Of the true No's, we make $23/9667 = 0.2\%$ errors; of the true Yes's, we make $252/333 = 75.7\%$ errors!

Concepts

		True condition				
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$	
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$	
	Predicted condition negative	**False negative**, Type II error	**True negative**	False omission rate (FOR) = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Predicted condition negative}}$	
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR−}}$	F_1 score = $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$	Negative likelihood ratio (LR−) = $\frac{\text{FNR}}{\text{TNR}}$		

Types of errors

False positive rate: The fraction of negative examples that are classified as positive — 0.2% in example.

False negative rate: The fraction of positive examples that are classified as negative —75.7% in example.

We produced this table by classifying to class Yes if

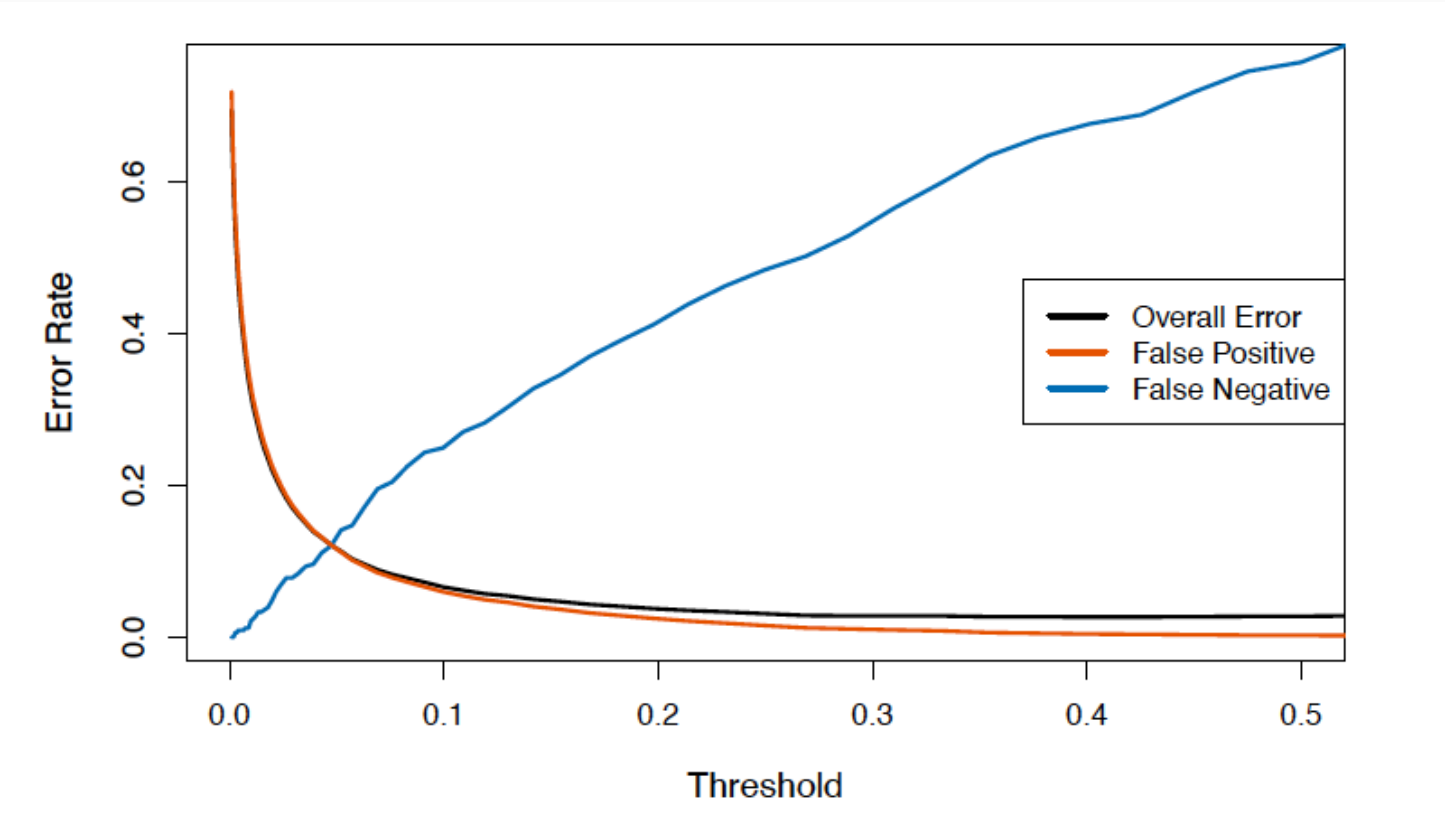
$$\hat{Pr}(\text{Default} = \text{Yes}|\text{Balance, Student}) \geq 0.5$$

We can change the two error rates by changing the threshold from 0.5 to some other value in $[0, 1]$:

$$\hat{Pr}(\text{Default} = \text{Yes}|\text{Balance, Student}) \geq \text{threshold},$$

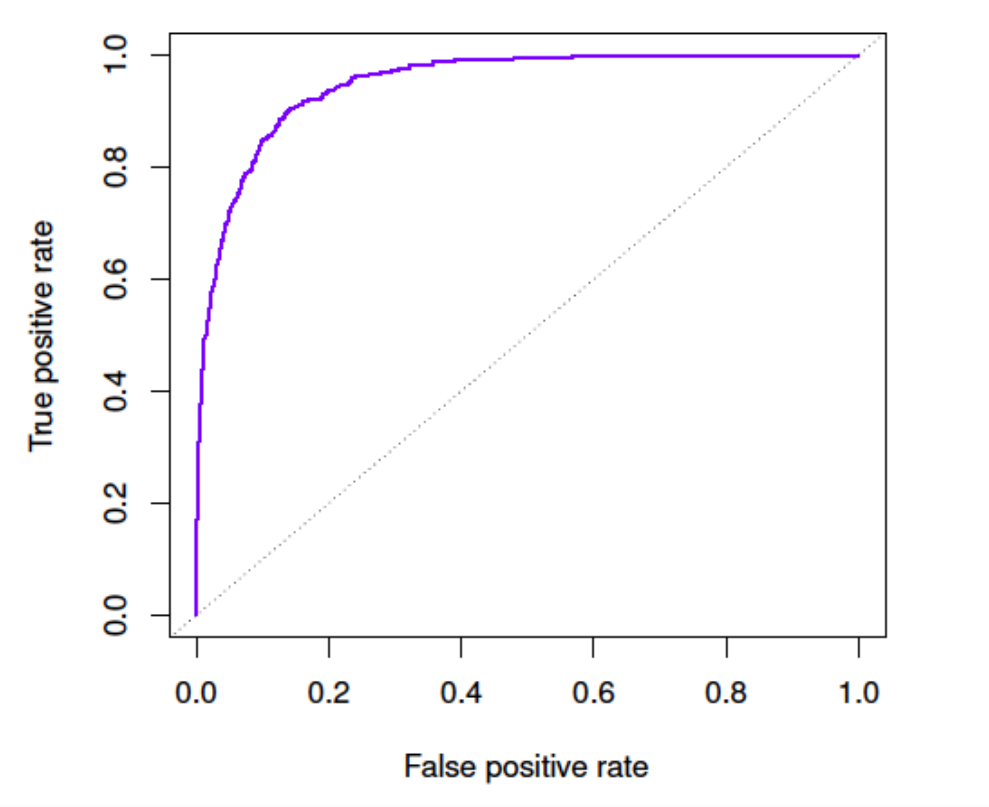
and vary threshold.

Varying the threshold



In order to reduce the false negative rate, we may want to reduce the threshold to 0.1 or less.

ROC curve



The ROC plot displays both simultaneously.

Sometimes we use the AUC or area under the curve to summarize the overall performance. Higher AUC is better.

Quadratic discriminant analysis (QDA)

Assuming $\mathbf{x}\,|y = k \sim N_p(\mu_k, \Sigma_k)$, then

$$\delta_k(\mathbf{x}) = \log(\pi_k) - (1/2)\log|\Sigma_k| - (1/2)(\mathbf{x} - \mu_k)^T \Sigma_k^{-1}(\mathbf{x} - \mu_k).$$

- The quadratic term of $\mathbf{x}$ now is necessary
- $\hat{\pi}_k = n_k/n$
- μ_k is estimated by the centroid in each class k
- Σ_k is estimated by sample covariance matrix in each class

Regularized discriminant analysis

- Regularized QDA $\hat{\Sigma}_k(\alpha) = \alpha\hat{\Sigma}_k + (1-\alpha)\hat{\Sigma}$
- Regularized LDA $\hat{\Sigma}(\gamma) = \gamma\hat{\Sigma} + (1-\gamma)\hat{\sigma}I_p$
- Together $\rightarrow \hat{\Sigma}(\alpha, \gamma)$
- Could use $\hat{\Sigma}(\gamma) = \gamma\hat{\Sigma} + (1-\gamma)\text{diag}(\hat{\Sigma})$

Reduced rank LDA

- Let $\widehat{\Sigma} = \mathbf{U}\,\mathbf{D}\,\mathbf{U}^T$ (eigen-decomposition)
- Let $\mathbf{x}^* = \mathbf{D}^{-1/2}\,\mathbf{U}^T\,\mathbf{x} = \hat{\Sigma}^{-1/2}\,\mathbf{x}$
- Let $\hat{\mu}_k^* = \mathbf{D}^{-1/2}\,\mathbf{U}^T\,\hat{\mu}_k$
- Then the discriminant function of LDA becomes

$$\delta_k(\mathbf{x}) = \frac{1}{2}\|\mathbf{x}^* - \hat{\mu}_k^*\|^2 - \log \hat{\pi}_k$$

- If $p > K$, we can project data onto the $(K-1)$ dimensional space spanned by $\hat{\mu}_k^*$, without losing any information.

We can project onto even lower dimensions, using the principal components of $\hat{\mu}_k^*$:

- Let $\mathbf{M}$ be the $K \times p$ matrix of centroids, $\mathbf{W} = \hat{\Sigma}$, and

$$\mathbf{M}^* = \mathbf{M}\,\mathbf{W}^{-1/2}$$

- Let $\mathbf{B}^*$ be the covariance matrix of $\mathbf{M}^*$, and its SVD decomposition is

$$\mathbf{B}^* = \mathbf{V}^* \mathbf{D}_B (\mathbf{V}^*)^T$$

- $\mathbf{z}_l = \mathbf{v}_l^T \mathbf{X}$ is the l-th discriminant (or canonical) variable, with $\mathbf{v}_l = \mathbf{W}^{-1/2}\,\mathbf{v}_l^*$

Example: Vowel training data

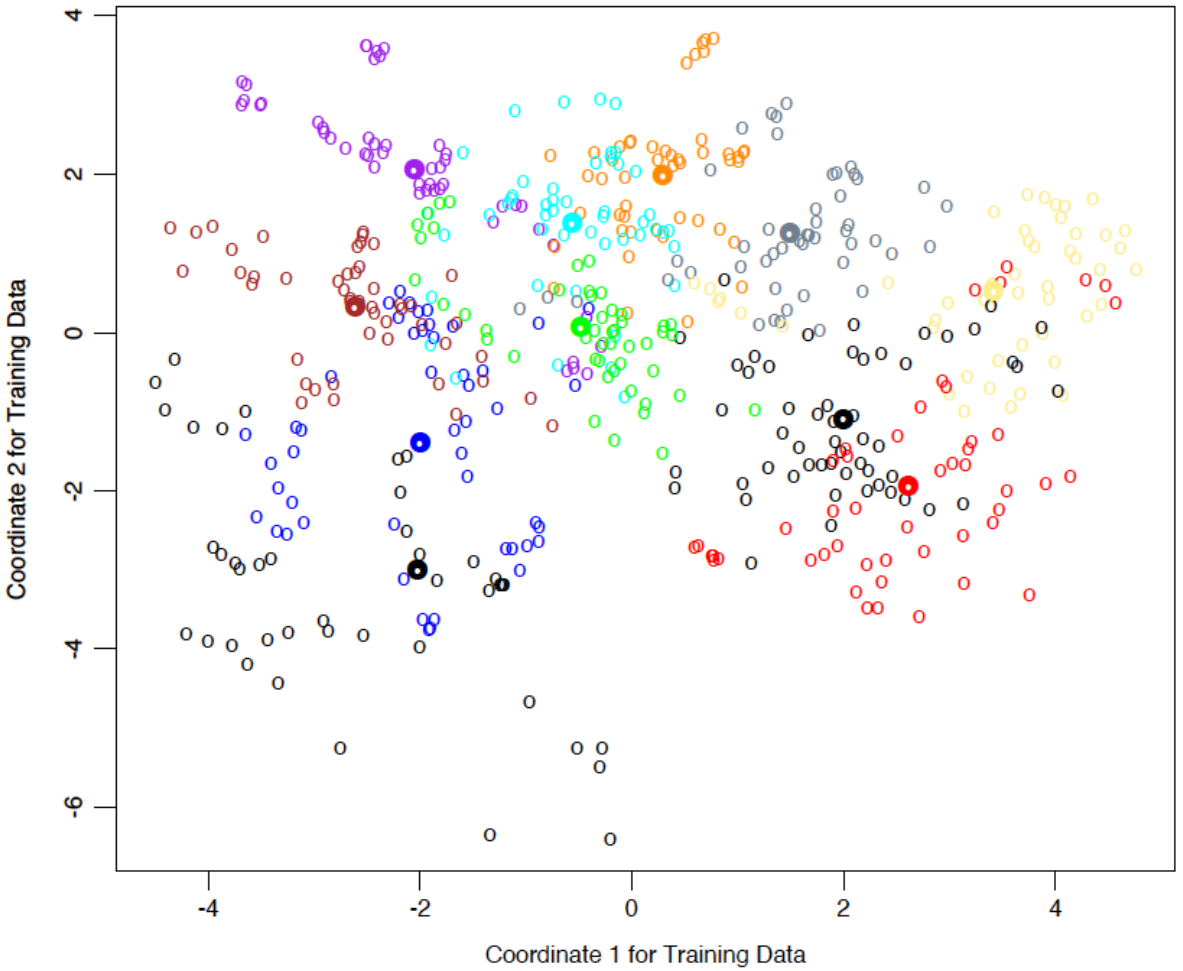
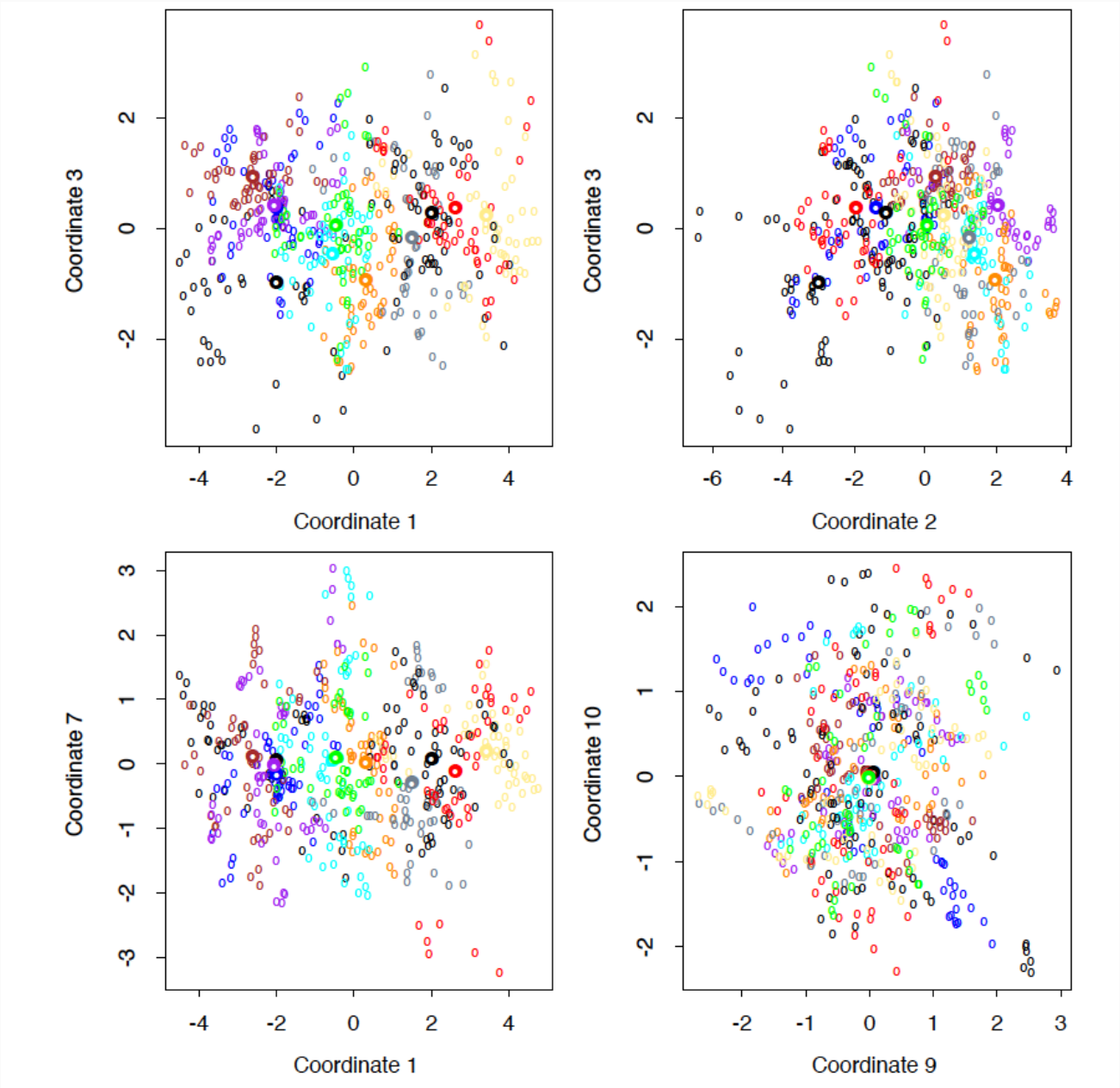


FIGURE 4.4. *A two-dimensional plot of the vowel training data. There are eleven classes with $X \in \mathbb{R}^{10}$, and this is the best view in terms of a LDA model (Section 4.3.3). The heavy circles are the projected mean vectors for each class. The class overlap is considerable.*

Example: Vowel training data



Example: Vowel training data

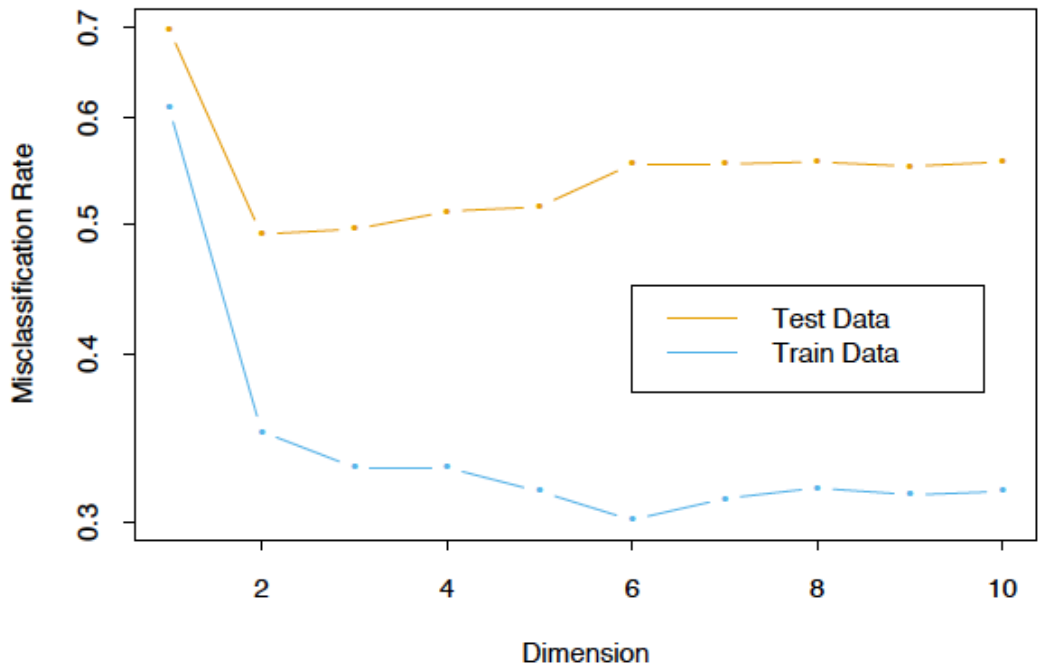


FIGURE 4.10. *Training and test error rates for the vowel data, as a function of the dimension of the discriminant subspace. In this case the best error rate is for dimension 2. Figure 4.11 shows the decision boundaries in this space.*

Example: Vowel training data

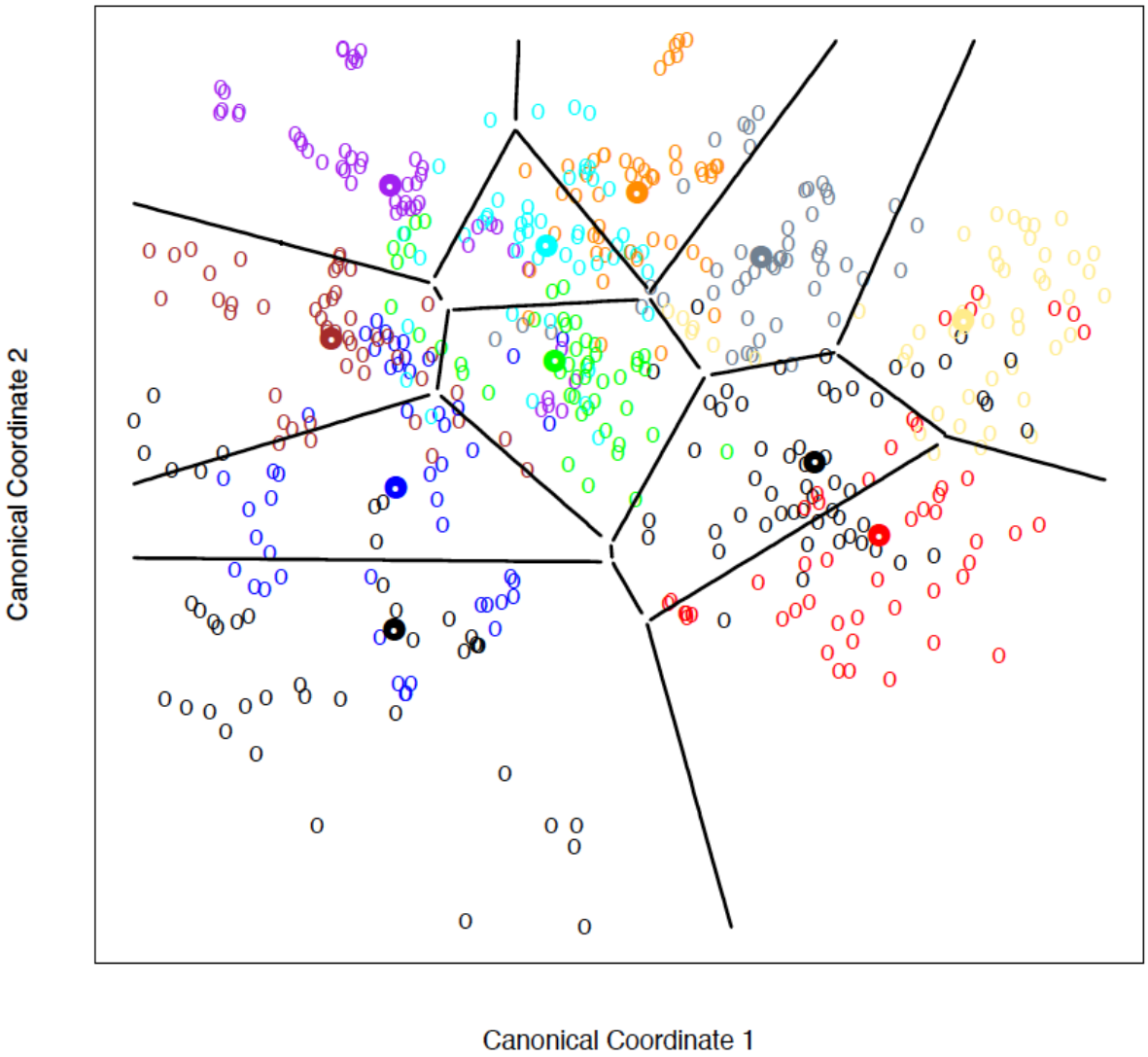


FIGURE 4.11. *Decision boundaries for the vowel training data, in the two-dimensional subspace spanned by the first two canonical variates. Note that in any higher-dimensional subspace, the decision boundaries are higher-dimensional affine planes, and could not be represented as lines.*

Fisher's formulation of discriminant analysis

This procedure can also be understood as finding $z = a^T \mathbf{x}$ so that the between-class variance is maximized relative to the within-class variance; that is,

$$\max_a \frac{a^T \mathbf{B}\, a}{a^T \mathbf{W}\, a},$$

or equivalently,

$$\max_a \; a^T \mathbf{B}\, a \;\; \text{s.t. } a^T \mathbf{W}\, a = 1,$$

where W is the common covariance matrix (within-class covariance) and B is the covariance matrix of the class centroid (between-class covariance). Note $B + W = T$, where T is the total covariance matrix of X, ignoring class information.

- Simple algebra implies that the solution is given by the first eigenvector of $\mathbf{W}^{-1}\mathbf{B}$.
- Given a_1, repeat the procedure to find a_2 with additional constraint $a_1^T a_2 = 0$.

Naive Bayes

Assumes features are independent in each class.

Useful when p is large, and so multivariate methods like QDA and even LDA break down.

- Gaussian naive Bayes assumes each Σ_k is diagonal:

$$\delta_k(x) \propto \log[\pi_k \prod_{j=1}^{p} f_{kj}(x_j)] = -\frac{1}{2} \sum_{j=1}^{p} \frac{(x_j - \mu_{kj})^2}{\sigma_{kj}^2} + \log \pi_k$$

- can use for mixed feature vectors (qualitative and quantitative). If X_j is qualitative, replace $f_{kj}(x_j)$ with probability mass function (histogram) over discrete categories.

Despite strong assumptions, naive Bayes often produces good classification results.

Logistic Regression

Logistic regression

Let's write $p(X) = Pr(Y = 1|X)$ for short and consider using balance to predict default. Logistic regression uses the form

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}.$$

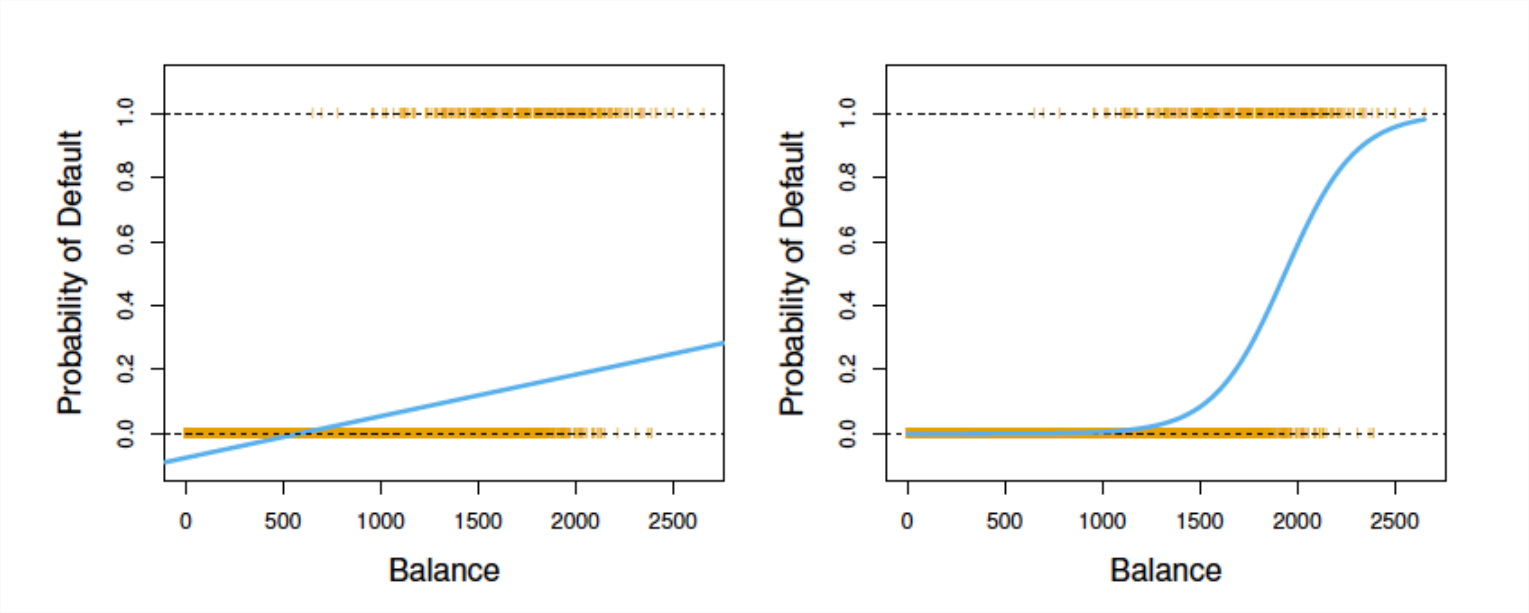
It is easy to see that no matter what values β_0, β_1, X take, $p(X)$ will have values between 0 and 1.

A big of rearrangement gives

$$\log(\frac{p(X)}{1 - p(X)}) = \beta_0 + \beta_1 X$$

This monotone transformation is called the log odds or logit transformation of $p(X)$.

Linear vs. Logistic regression



Logistic regression ensures that our estimate for $p(X)$ lies between 0 and 1.

Logistic regression

Linear logistic regression assumes that in binary classification with $y \in \{0, 1\}$,

$$\text{logit}(p(\mathbf{x})) \equiv \log \frac{p(\mathbf{x})}{1 - p(\mathbf{x})} = \mathbf{x}^T \beta,$$

where $p(\mathbf{x}) = P(y = 1 | \mathbf{x})$ and β is the unknown parameter

Its log-likelihood is

$$l(\beta) = \sum_{i=1}^{n} \Big(y_i \log p_i + (1 - y_i) \log(1 - p_i) \Big),$$

where $p_i = p(\mathbf{x}_i) = (1 + \exp(\mathbf{x}_i^T \beta))^{-1}$

Logistic regression

To maximize $l(\beta)$, simple algebra yields that

$$\begin{aligned} l(\beta) &= \sum_{i=1}^{n}(y_i(\mathbf{x}^T\beta) - \log(1+\exp(\mathbf{x}^T\beta))), \\ l'(\beta) &= \sum_{i=1}^{n}(y_i - p_i)\mathbf{x}_i = \mathbf{X}^T(\mathbf{y} - \mathbf{p}), \\ l''(\beta) &= -\sum_{i=1}^{n} p_i(1-p_i)\mathbf{x}_i\mathbf{x}_i^T = -\mathbf{X}^T\mathbf{W}\mathbf{X}, \end{aligned}$$

where $\mathbf{W} = \text{diag}\{p_1(1-p_1), \ldots, p_n(1-p_n)\}$.

Iteratively reweighted least square

Newton-Raphson method updates β iteratively,

$$\begin{aligned}\beta^{new} &= \beta^{old} + (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1}(\mathbf{X}^T(\mathbf{y} - \mathbf{p})) \\ &= (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1}\mathbf{X}^T \mathbf{W}(\mathbf{X}\beta^{old} + \mathbf{W}^{-1}(\mathbf{y} - \mathbf{p})) \\ &= (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1}\mathbf{X}^T \mathbf{W}\mathbf{y}^{new}.\end{aligned}$$

In the last two lines we have re-expressed the Newton-Raphson step as a weighted least squares step, with the new response

$$\mathbf{y}^{new} = \mathbf{X}\beta^{old} + \mathbf{W}^{-1}(\mathbf{y} - \mathbf{p}).$$

$$\beta^{new} \longleftarrow \arg\min_{\beta}(\mathbf{y}^{new} - \mathbf{X}\,\beta)^T \mathbf{W}(\mathbf{y}^{new} - \mathbf{X}\,\beta)$$

Example: South African heart disease

- 160 cases of MI (myocardial infarction) and 302 controls (all males in age range 15-64), from Western Cape, South Africa in early 80s.
- Overall prevalence very high in this region: 5.1%.
- Measurements on seven predictors (risk factors), shown in scatterplot matrix.
- Goal is to identify relative strengths and directions of risk factors.
- This was part of an intervention study aimed at educating the public on healthier diets.

Example: South African heart disease

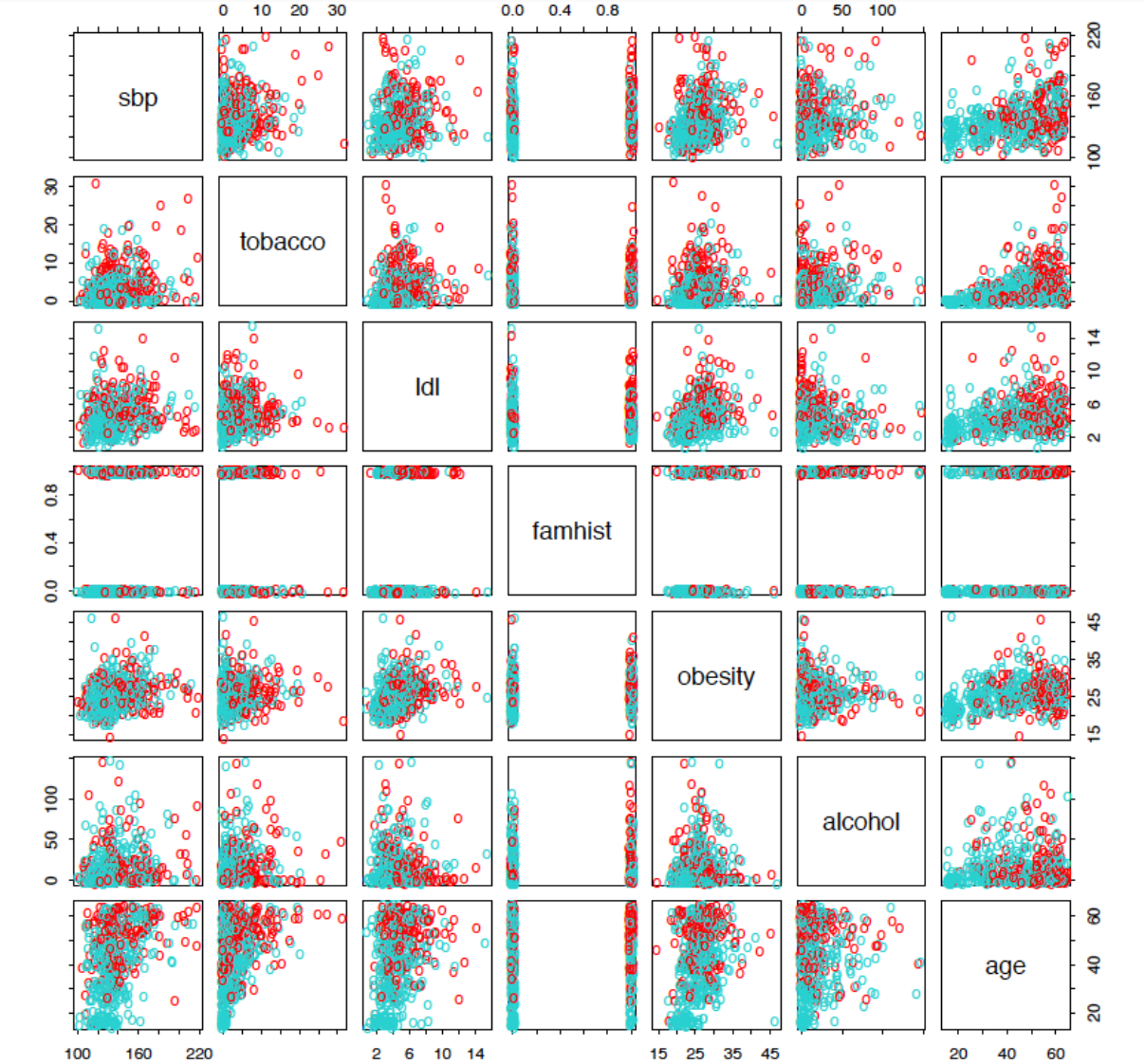


FIGURE 4.12. *A scatterplot matrix of the South African heart disease data. Each plot shows a pair of risk factors, and the cases and controls are color coded (red is a case). The variable* family history of heart disease (`famhist`) *is binary (yes or no).*

Example: South African heart disease

Table 1: Logistic regression fit to the South African heart disease data.

	Coef.	Std. Err.	Z score
(Intercept)	-4.130	0.964	-4.285
sbp	0.006	0.006	1.023
tobacco	0.080	0.026	3.034
ldl	0.185	0.057	3.219
famhist	0.939	0.225	4.178
obesity	-0.035	0.029	1.187
alcohol	0.001	0.004	0.136
age	0.043	0.010	4.184

Note: a stepwise logistic regression fit is given in Table 4.3. A likelihood ratio test can be employed in each step.

TABLE 4.3. *Results from stepwise logistic regression fit to South African heart disease data.*

	Coefficient	Std. Error	Z score
`(Intercept)`	−4.204	0.498	−8.45
`tobacco`	0.081	0.026	3.16
`ldl`	0.168	0.054	3.09
`famhist`	0.924	0.223	4.14
`age`	0.044	0.010	4.52

Case-control sampling and logistic regression

- In South African data, there are 160 cases, 302 controls — $\tilde{\pi} = 0.35$ are cases. Yet the prevalence of MI in this region is $\pi = 0.05$.
- With case-control samples, we can estimate the regression parameters β_j accurately (if our model is correct); the constant term β_0 is incorrect.
- We can correct the estimated intercept by a simple transformation

$$\hat{\beta}_0^* = \hat{\beta}_0 + \log \frac{\pi}{1-\pi} - \log \frac{\tilde{\pi}}{1-\tilde{\pi}}$$

- Often cases are rare and we take them all; up to five times that number of controls is sufficient.

Other extensions

- Nonlinear logistic regression assumes a nonlinear or even nonparametric form of logit($p(\mathbf{x})$).
- Multiclass logistic regression assumes

$$\begin{aligned}
\log \frac{p_1(\mathbf{x})}{p_K(\mathbf{x})} &= \eta_1(\mathbf{x}) = \mathbf{x}^T \beta_1, \\
\log \frac{p_2(\mathbf{x})}{p_K(\mathbf{x})} &= \eta_2(\mathbf{x}) = \mathbf{x}^T \beta_2, \\
&\vdots \\
\log \frac{p_{K-1}(\mathbf{x})}{p_K(\mathbf{x})} &= \eta_{K-1}(\mathbf{x}) = \mathbf{x}^T \beta_{K-1}.
\end{aligned}$$

- L_1-norm regularization as in Lasso can be used for variable selection and shrinkage in logistic regression.

Logistic regression vs. LDA

For a two-class problem, one can show that for LDA

$$\log(\frac{p_1(x)}{1-p_1(x)}) = \log(\frac{p_1(x)}{p_2(x)}) = c_0 + c_1 x_1 + \cdots + c_p x_p$$

So it has the same form as logistic regression.

The difference is in how the parameters are estimated.

- Logistic regression uses the conditional likelihood based on $Pr(Y|X)$ (known as discriminative learning)
- LDA uses the full likelihood based on $Pr(X, Y)$ (known as generative learning)
- Despite these differences, in practice, the results are often very similar.

Footnote: logistic regression can also fit quadratic boundaries like QDA, by explicitly including quadratic terms in the model.

Summary

- Logistic regression is very popular for classification, especially when $K = 2$.
- LDA is useful when n is small, or the classes are well separated, and Gaussian assumptions are reasonable. Also when $K > 2$.
- Naive Bayes is useful when p is very large.
- See Section 4.5 for some comparisons of logistic regression, LDA and KNN.